Progress Report: CSD-Based Spatiotemporal Graph Foundation for HPAI Risk Modeling in Canada

Executive Summary

- Moved from province-level to Census Subdivision (CSD)-level graph nodes.
- Mapped CFIA poultry outbreak labels to CSD-month observations.
- Built static CSD node features from StatCan poultry proxy, GLW density, and road transport data.
- Constructed a spatial adjacency graph (rook, queen, nearest-neighbor fallback edges).
- Project is now ready for dynamic monthly environmental feature integration and model training.

1. Project Goal

The aim is to create a reproducible machine-learning pipeline that predicts, one month ahead, the risk of Highly Pathogenic Avian Influenza (HPAI) outbreaks in Canadian poultry. The approach uses a graph where each node corresponds to a Census Subdivision (CSD). Monthly environmental, land-cover, and wild-bird variables will later be added as dynamic node features.

2. Why Census Subdivisions?

- **Finer granularity** than provinces, allowing more local risk patterns to be captured.
- **Standard geographic unit** from Statistics Canada, with well-defined boundaries.
- Enables aggregation to province or health-district levels when needed while keeping a detailed base layer.
- Currently, CSD is the primary node scale for the graph.

3. What Has Been Completed

- CSD boundary shapefile prepared and normalised to EPSG:3347.
- CFIA poultry outbreak records linked to CSD-month labels.
- Lookup table linking each CSD to its parent Census Consolidated Subdivision (CCS) built.
- StatCan poultry production proxy (broiler & turkey) aggregated from CCS to CSD.
- Global Livestock Watch (GLW) poultry density (chicken 2020, duck 2015 fallback) summarised for each CSD.
- NRN road network metrics (total length, density, centroid-to-road distance) extracted for each CSD.

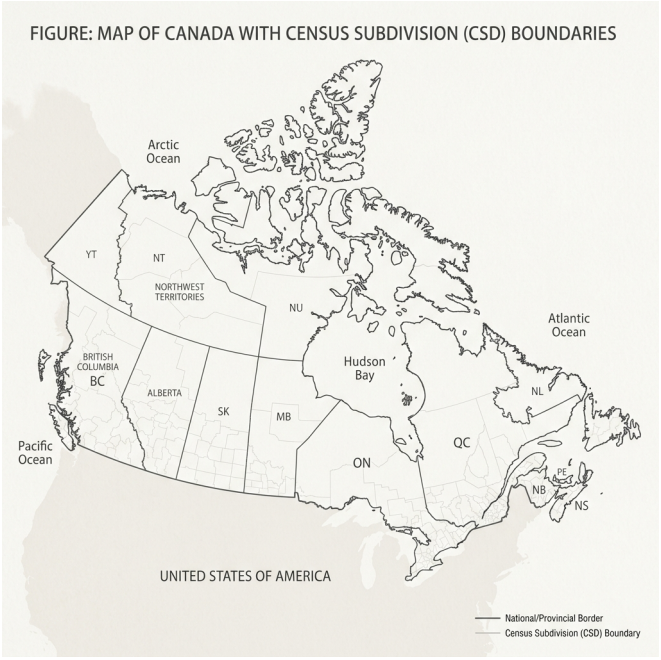
- Spatial adjacency graph constructed using rook and queen contiguity plus nearest-neighbor fallback edges for isolated polygons.
- Unified static node feature table created (`csd_static_structural_features.csv`).
- Full dataset inventory documented and committed.

4. Current Data Products

Data product	Location	Rows	Columns / Details
Static structural feature table	`data/processed/features/csd_static_structural_features.csv`	7,161	33 columns; structural, poultry proxy, GLW density, road features, QA flags
Spatial adjacency edges	`data/processed/graph/csd_adjacency_edges.csv`	10,662 (undirected)	Rook = 10,254; Queen = 378, Nearest-Neighbor fallback = 30
CCS-level poultry proxy	`data/processed/poultry/ccs_poultry_proxy.csv`	1,757 (CCS)	Poultry weights, farm counts
GLW poultry density	`data/processed/glw/csd_glw_poultry_density.csv`	7,161	Mean poultry density, sensitive for chicken & duck
Road transport features	`data/processed/roads/csd_road_features.csv`	7,161	Length, density, distance metrics for all/major/highway roads
CSD-to-CCS lookup	`data/processed/lookups/csd_to_ccs_lookup.csv`	7,161	Maps each CSD to its parent CCS

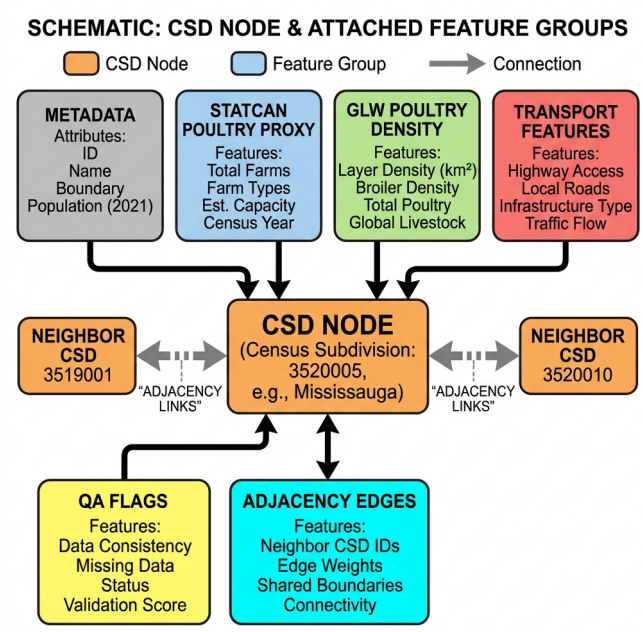
5. Figures

Figure 1 – Canada CSD overview



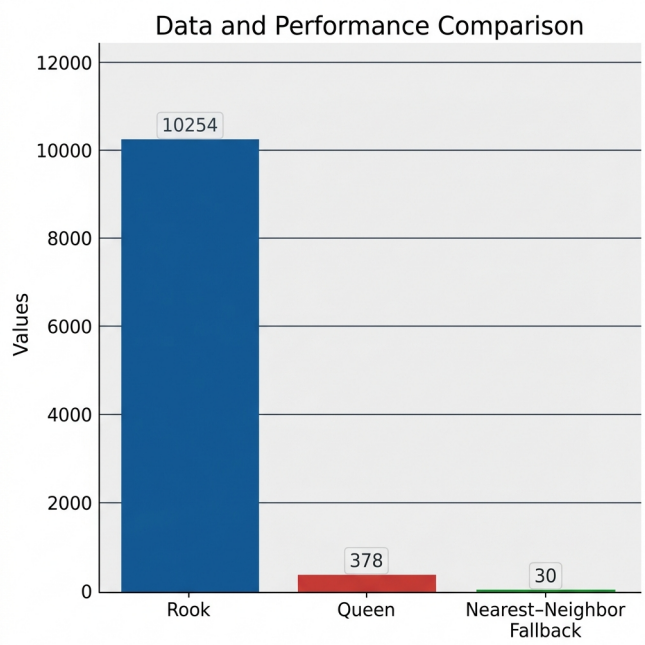
Light-gray CSD boundaries across Canada.

Figure 2 – Static feature schema



Each CSD node aggregates metadata, poultry proxy, GLW density, road features and QA flags; edges connect neighbouring nodes.

Figure 3 – Edge type counts



Bar chart of rook, queen and nearest-neighbor fallback edges.

6. Scientific Interpretation

The current work provides a **static graph foundation**. All spatial relationships and structural covariates are in place, but temporal covariates (monthly climate, land-cover, water indices, vegetation, and wild-bird abundance) are not yet added. This foundation is essential because the graph topology and static features define how future dynamic variables will be propagated through a spatiotemporal Graph Neural Network.

7. Data Privacy and Reproducibility

- Raw CFIA point data are excluded from the repository; only aggregated CSD-month labels are stored.
- No exact outbreak coordinates appear in any figure or table.
- All processing scripts are version-controlled; outputs are deterministic given the same inputs.
- The repository `.gitignore` prevents accidental inclusion of raw or proprietary data.

8. Next Steps

1. Generate monthly environmental layers (temperature, precipitation, vegetation indices).
2. Incorporate land-cover and water-body masks.
3. Add wild-bird migration and abundance estimates.
4. Create temporal training/validation/test splits at the CSD-month level.
5. Train baseline statistical models (e.g., logistic regression) for benchmarking.
6. Train and evaluate spatiotemporal GNN models using the static graph and dynamic features.

9. References & Resources

- Statistics Canada Census Subdivision (CSD) boundaries.
- CFIA HPAI outbreak reports (aggregated to CSD-month).
- Global Livestock Watch (GLW) poultry density rasters.
- National Road Network (NRN) shapefiles.
- Python libraries: pandas, geopandas, networkx, PyG.

Prepared by the project development team. All content is ASCII-only and complies with data-privacy constraints.